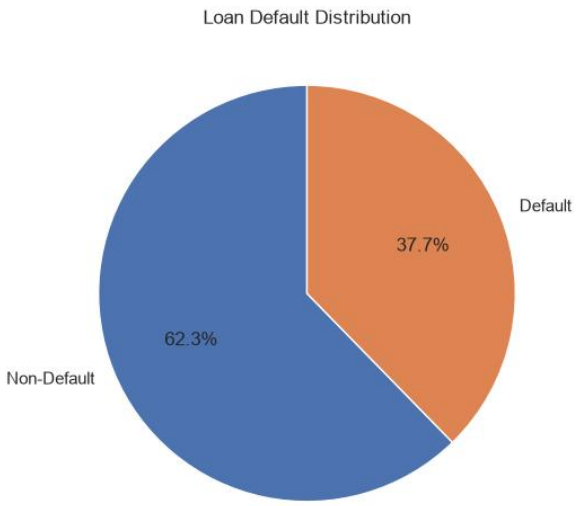

Loan Default Distribution
Default
37.7%
62.3%
Non-Default

Fig 1

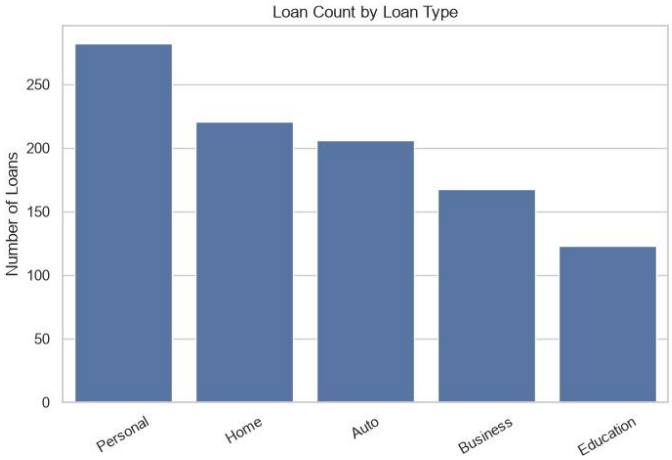

Loan Count by Loan Type
Number of Loans
250
200
150
100
50
0
Personal
Home
Auto
Business
Education

Fig 2

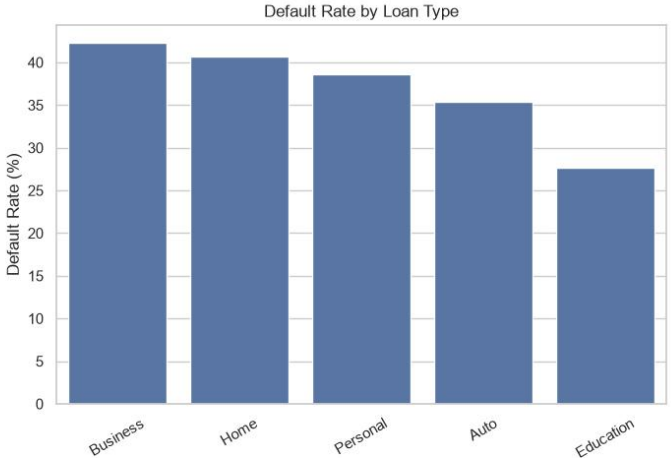

Default Rate by Loan Type
Default Rate (%)
40
35
30
25
20
15
10
5
0
Business
Home
Personal
Auto
Education

Fig 3

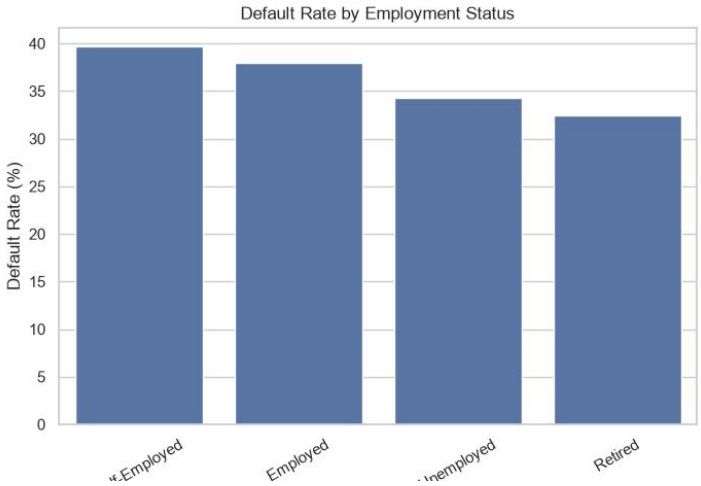


Fig 4

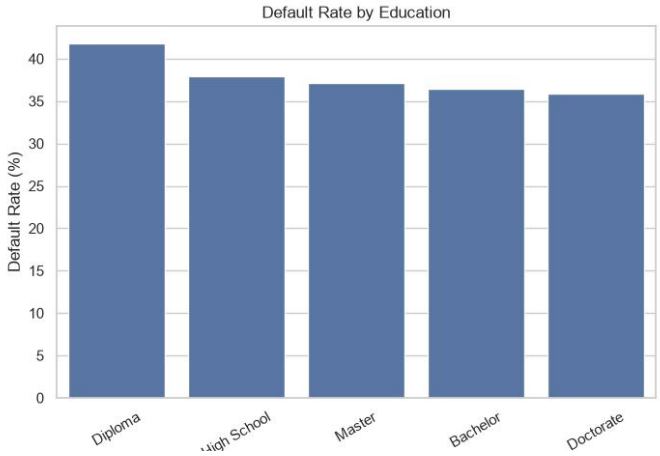


Fig 5

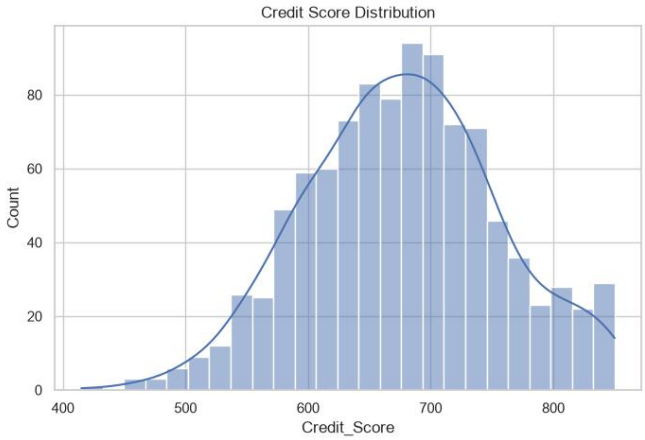


Fig 6

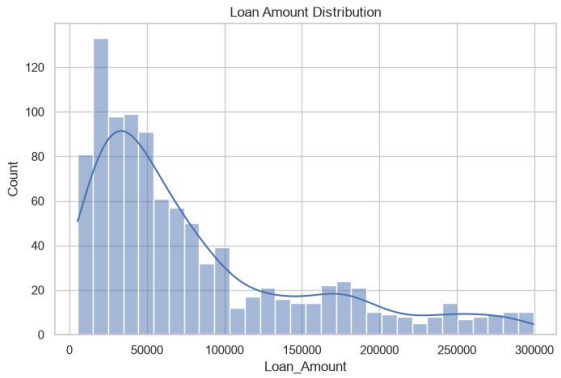


Fig 7

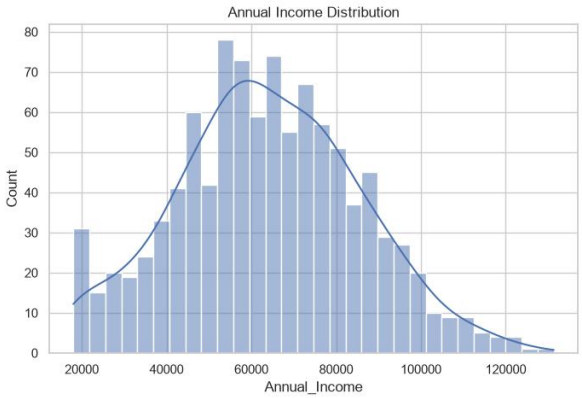


Fig 8

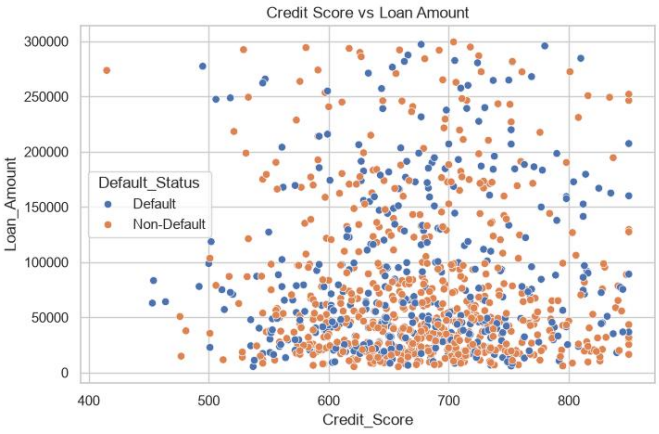


Fig 9

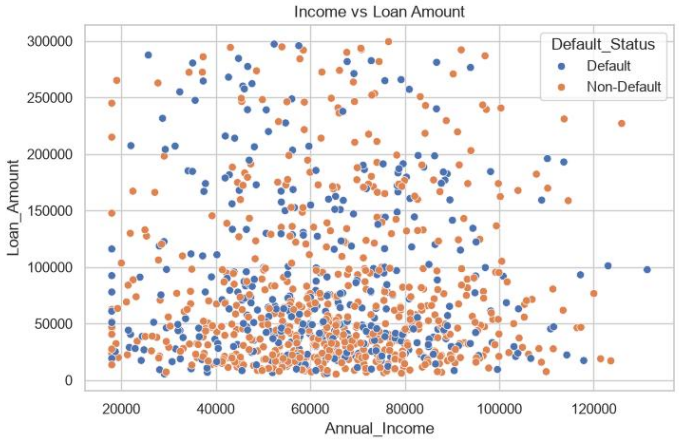


Fig 10

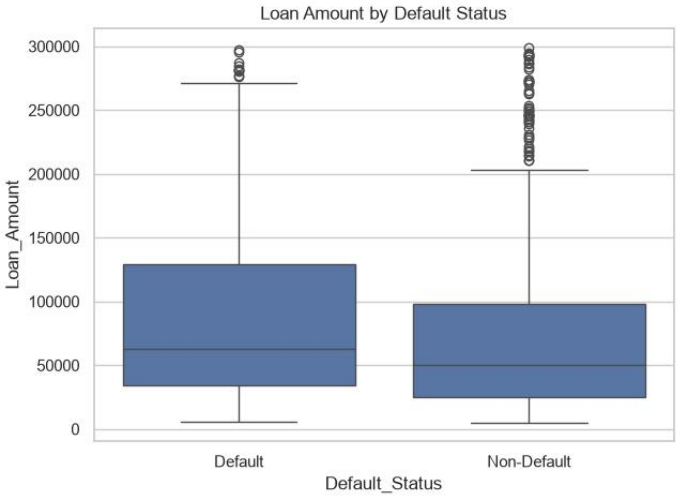


Fig 11

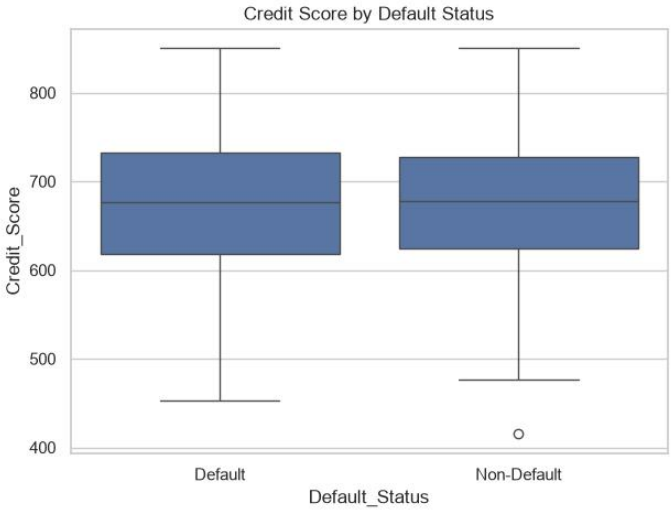


Fig 12

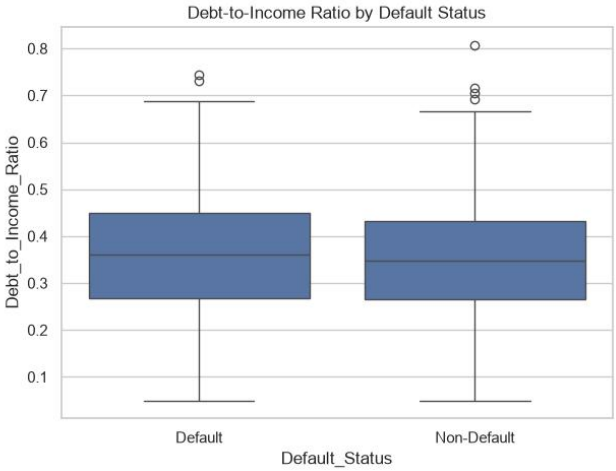


Fig 13

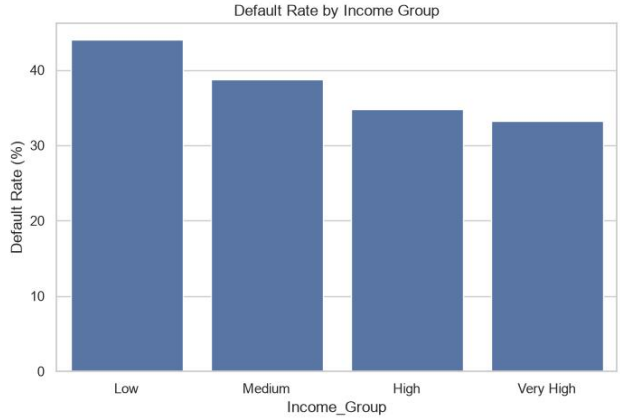


Fig 14

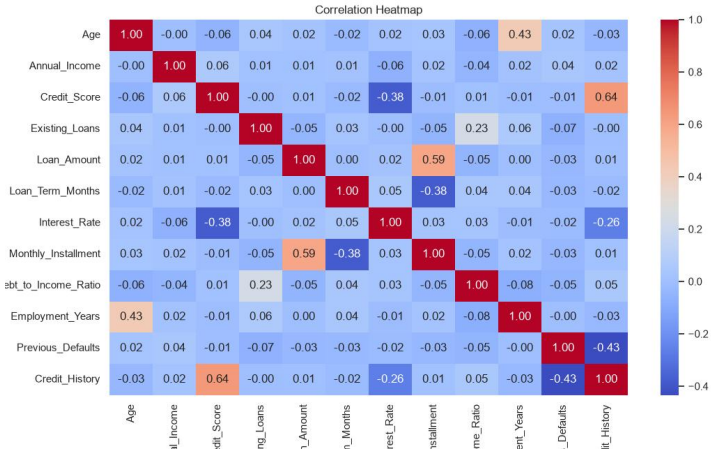


Fig 15

BUSINESS INSIGHTS

1. What is the overall loan default rate?

You already created the DAX measure:

```
Default Rate % =
DIVIDE(
    [Defaulted Loans],
    [Total Loans],
    0
)
```

Dataset has:

- Total Loans = **1,000**
- Defaulted Loans = **377**

Therefore:

Overall Loan Default Rate = 37.70%

How to show it in Power BI

Use your **Default Rate % Card**.

It should display:**37.70%**

2. Which loan type has the highest default rate?

Already created:Visual

Default Rate by Loan Type — Bar Chart

Fields:

Y-axis → Loan_Type
X-axis → Default Rate %

3. Which region has the highest default rate?

Default Rate by Region

If you don't already have this visual, create a **Bar Chart**.

Fields

Y-axis → Region
X-axis → Default Rate %

4. Does credit score appear associated with loan default?

Default Rate by Credit Score Group

I created:

Poor
Fair
Good
Very Good
Excellent

Power BI visual

X-axis → Credit_Score_Group
Y-axis → Default Rate %

What to look for

Compare the heights of the columns.

If you see something like:

Poor
Fair
Good
Very Good
Excellent



5. Does income appear associated with default?

Default Rate by Income Group

Your groups are:

Low
Medium
High
Very High

Fields

Y-axis → Income_Group
X-axis → Default Rate %

What to look for

Compare the four bars.

For example:

Low
Medium
High
Very High



6. Are higher debt-to-income ratios associated with default?

Debt_to_Income_Ratio

Default Rate by DTI Group

Low DTI
Medium DTI
High DTI



then higher DTI is associated with higher default risk.

7. Are previous defaults associated with current default?

This is one of the strongest risk variables to investigate.

My dataset has:

Previous_Defaults

You can compare:

Previous_Defaults
↓
Default_Status

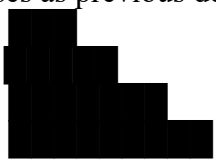
Use a **Clustered column chart**.

X-axis → Previous_Defaults
Y-axis → Default Rate %

If Previous_Defaults is numeric, you can compare categories such as:

0 previous defaults
1 previous default
2 previous defaults

If default rate increases as previous defaults increase:
0 previous defaults
1 previous default
2 previous defaults
3 previous defaults



then there is a strong relationship.

8. Which customer segment has the highest default rate?

There isn't just one possible definition of "customer segment."

For my project, you can compare segments based on:

- Employment Status
- Credit Score Group
- Income Group
- Education
- Gender
- Loan Type
- Region

Use charts and identify the group with the highest default rate.

For example:

Credit Score Group

Poor
Fair
Good
Very Good
Excellent

or:

Employment Status

Employed
Self-Employed
Unemployed
Retired

9. Which factors appear associated with loan default?

This is where we bring all your analysis together.

important factors are:

Credit-related factors

- Credit Score
- Credit History
- Previous Defaults

Financial factors

- Annual Income
- Debt-to-Income Ratio
- Loan Amount
- Interest Rate
- Monthly Installment

Customer factors

- Age
- Education
- Employment Status
- Marital Status
- Dependents

Loan factors

- Loan Type
- Loan Term
- Property Ownership
- Region

10. What patterns can be observed between loan amount and default status?

Income vs Loan Amount — Scatter Chart

Recommended fields:

X-axis → Annual_Income
Y-axis → Loan_Amount
Legend → Default_Status
Details → Customer_ID

We can also create:

Average Loan Amount by Default Status

Recommended additional visual

Create a **Column Chart**:

X-axis → Default_Status
Y-axis → Average Loan Amount

Then compare:

Default [redacted]
Non-Default [redacted]

11. What financial factors should an organization monitor?

For a bank or lending organization, I would recommend monitoring:

1. Credit Score

Lower credit scores may indicate greater risk.

2. Debt-to-Income Ratio

High DTI can indicate greater repayment pressure.

3. Annual Income

Income indicates the borrower's repayment capacity.

4. Previous Defaults

Previous repayment problems are an important risk signal.

5. Loan Amount

Large exposures can increase potential financial loss.

6. Interest Rate

Higher interest rates can increase repayment burden.

7. Monthly Installment

The installment should be reasonable relative to income.

8. Credit History

A strong credit history can indicate better repayment behavior.

12. Write 5–10 Key Business Insights

You can use the following as the **Business Insights** section of your project report. Replace the bracketed items with the actual highest categories from your Power BI charts.

Key Business Insights

1. Overall Default Rate

The overall loan default rate is **37.70%**, indicating that a significant proportion of loans in the analyzed dataset are classified as defaulted.

2. Loan Type Risk

The loan type with the highest default rate is **[insert loan type from your chart]**. This category should receive additional attention during credit-risk assessment.

3. Regional Risk

[Insert region] records the highest default rate among the analyzed regions, suggesting that regional differences may be relevant to credit-risk monitoring.

4. Credit Score

Credit score appears to be associated with default. Lower credit-score groups generally represent higher-risk borrowers, while stronger credit scores are associated with lower default risk.

5. Income

Default rates differ across income groups. Income therefore appears to be a useful factor for understanding differences in repayment risk.

6. Debt-to-Income Ratio

Higher debt-to-income ratios are associated with greater financial pressure and may correspond to higher default risk.

7. Previous Defaults

Customers with previous default records represent an important high-risk segment because past repayment problems may be associated with current default.

8. Loan Amount

Loan amount should be considered together with income and debt burden. Larger loan exposure can increase the financial impact of default.

9. Multiple Risk Factors

Loan default is not determined by one factor alone. Credit score, previous defaults, DTI, income, employment status, loan type, and loan amount collectively provide useful information about borrower risk.

10. Risk Monitoring

Financial institutions should use a combination of **credit, income, debt, repayment-history, and loan characteristics** when evaluating customers rather than relying on a single indicator.